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HIERARCHICAL CONFORMING FINITE ELEMENT METHODS FOR THE BIHARMONIC EQUATION*

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Abstract. The paper deals with hierarchical bases in spaces of conforming C^1 elements in connection with the approximate solution of the biharmonic equation

$$\Delta^2 u = f \quad \text{in } \Omega, \qquad u = \frac{\partial u}{\partial n} = 0 \quad \text{on } \partial\Omega,$$

on a plane polygonal domain Ω. Two different composite finite elements are studied: piecewise quadratic Powell-Sabin elements and piecewise cubic elements of Clough-Tocher type.

The main result are estimates for the condition numbers of the corresponding discretization matrices that show that a conjugate gradient method applied to the hierarchical discretization (the so-called hierarchical multilevel method) will yield suboptimal convergence rates in comparison with standard multigrid schemes.

Key words. hierarchical multilevel methods, C^1 finite elements, preconditioned conjugate gradient method, biharmonic equation

AMS(MOS) subject classifications. 65F10, 65F35, 65N30

Introduction. In recent years, preconditioners for solving the finite element equations for second-order elliptic problems by conjugate gradient methods have been constructed by introducing certain hierarchical multilevel splittings into the finite element spaces (cf. [Y1]-[Y4], [B-Du-Y], [B-P-X], [X]). Though closely related to standard multigrid methods, the hierarchical method offers some specific features (e.g., less restrictive assumptions on the regularity of the solutions, the option to allow local refinements without destroying the theoretical properties of the method, and the possible parallelization of the computations) which makes it interesting for further investigation.

The aim of the present paper is to develop the theory of a hierarchical multilevel method for the biharmonic equation

$$\Delta^2 u = f \quad \text{in } \Omega, \quad u = \frac{\partial u}{\partial n} = 0 \quad \text{on } \partial\Omega, \quad f \in L_2(\Omega), \tag{0.1}$$

where $\Omega \subset \mathbb{R}^2$ is a polygonal domain. We restrict our attention to two different types of conforming C^1 elements for discretizing the variational problem

$$a(u, v) = (f, v) \quad \text{for all } v \in H_0^2(\Omega), \tag{0.2}$$

corresponding to (0.1): Powell-Sabin triangles and modified Clough-Tocher triangles (as usual, (f, v) denotes the scalar product in $L_2(\Omega)$, and the bilinear form $a: H_0^2 \times H_0^2 \to \mathbb{R}$ is defined as $a(u, v) = (\Delta u, \Delta v)$; cf. [Cia] for more background).

To keep the exposition as simple as possible we consider regular dyadic refinements of a given initial triangulation T_0 which leads to an increasing sequence of quasi-uniform triangulations and to corresponding finite element spaces $S_0 \subseteq S_1 \subseteq \cdots \subseteq S_i \subseteq \cdots$. Using the natural degrees of freedom associated with the above element types, the locally supported nodal basis functions $\{N_{i,j}\}$ for S_i and interpolation projections

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$I_i : C^1(\bar{\Omega}) \to S_i$ will be introduced. The crucial point for the theory of the hierarchical multilevel method to be developed are estimates for the discrete norm

$$\|\bar{g}_j\|^2 = \|I_0 \bar{g}_j\|^2_{H^2} + \sum_{i=1}^{j} 2^{4i} \|I_i \bar{g}_j - I_{i-1} \bar{g}_j\|^2_{L_2}, \qquad \bar{g}_j \in S_j, \tag{0.3}$$

which also suggests the construction of the hierarchical basis in S_j. More precisely, the comparison of $\|\bar{g}_j\|$ and the H^2-norm of $\bar{g}_j$ will immediately yield estimates for the condition number of the linear system resulting from the discretization of (0.2) with respect to the hierarchical basis. It turns out that the condition numbers in the hierarchical case are significantly better than for the usual nodal discretization, which makes the hierarchical approach attractive for preconditioning finite element equations and solving them by the conjugate gradient method. We prove, for the above-mentioned element types, an $O(j^2)$ bound for the corresponding spectral condition numbers that leads to asymptotically suboptimal convergence rates when compared with a standard multigrid method.

The desired estimates for the discrete norm in (0.3) will be obtained within the framework of certain scales of approximation spaces $A^s_{p,q}(\{S_i\})$ the theory of which has previously been developed in [Os1] and [Os2] for subspaces $\{S_i\}$ of C^0 finite elements. Using rather standard techniques from approximation theory, in the present paper we consider the above-mentioned C^1 finite element subspaces and show, in particular, the norm equivalence

$$\|\bar{g}_j\|^2_{H^2} \approx \|\bar{g}_j\|^2_{A^2_{2,2}} \equiv \inf \sum_{i=0}^{j} 2^{4i} \|g_j\|^2_{L_2}, \qquad \bar{g}_j \in S_j, \tag{0.4}$$

where the infimum in (0.4) is taken with respect to all representations

$$\bar{g}_j = \sum_{i=0}^{j} g_i, \qquad g_i \in S_i, \quad i = 0, \cdots, j$$

(here and in the following, $A \approx B$ stands for a two-sided inequality $c \cdot A \leqq B \leqq C \cdot A$ with absolute constants c and C). Such norm characterizations seem to be very useful for analyzing multilevel-multigrid finite element methods. See [X], [Y4], and [Os1] for some more motivation to study $A^s_{p,q}$-norms (especially, in the case $p = q = 2$). For example, in [Os1] the problem of asymptotically optimal grid refinement has been treated using these function spaces.

Recently, we have learned about related research by Dörfler [Dö1], who presented a different but rather general approach to hierarchical multilevel methods for higher-order elliptic boundary value problems including both conforming and nonconforming finite element constructions. Though composite elements were not explicitly treated in [Dö1], a straightforward application of the arguments in [Dö1] to our situation leads to a condition number estimate of $O(j^3)$ (it is not quite clear whether the sharp $O(j^2)$ bound obtained in our paper can also be derived; cf. [Dö1, pp. 20–21]).

We finish this introduction by briefly indicating the contents of the remaining sections. In § 1, the triangulations and the finite element spaces together with the hierarchical basis systems are introduced for the element types mentioned above. The corresponding theory of $A^s_{p,q}$-spaces is developed in § 2. Section 3 is devoted to the discrete norm and condition number estimates. We finish with a short discussion of some implementational issues and further extensions.

1. Basic constructions. Throughout this paper, let Ω be a bounded connected polygonal domain in $\mathbb{R}^2$. Suppose that some initial triangulation of Ω is given and that

$$T_0 < T_1 < \cdots < T_i < \cdots, \qquad i = 0, 1, \cdots \tag{1.1}$$

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is the sequence of dyadically refined triangulations obtained by subdividing the triangles of T_i (which we call triangles of level i) into four equal subtriangles of level $i+1$, $i=0, 1, \cdots$ (cf. Fig. 1). The sequence $\{T_i\}$ satisfies the minimum angle condition (moreover, all triangles are similar to one of the triangles of level 0) and is quasi-uniform, i.e.,

$$C_1 \cdot 2^{-i} \leqq \operatorname{diam} \Delta \leqq C_2 \cdot 2^{-i}, \qquad \Delta \in T_i, \quad i=0, 1, \cdots, \tag{1.2}$$

with absolute constants C_1, C_2.

1.1. Powell–Sabin elements. Starting from (1.1) we introduce further substructures by subdividing each triangle of T_i into 12 subtriangles by joining the midpoints of the three edges with each other and with the opposite vertex (cf. Fig. 2).

The resulting sequence of triangulations $\{\mathcal{T}_i\}$ satisfies

$$\mathcal{T}_0 < \mathcal{T}_1 < \cdots < \mathcal{T}_i < \cdots, \qquad T_i < \mathcal{T}_i, \quad i=0, 1, \cdots, \tag{1.3}$$

$$h(\mathcal{T}_i) = \max \{\operatorname{diam} \Delta : \Delta \in \mathcal{T}_i\} \approx 2^{-i}, \qquad i \to \infty. \tag{1.4}$$

It is well known ([Po-Sa] or [Ch-He], [Fa]) that the spline space $S_i \equiv S_2^1(\mathcal{T}_i)$ of piecewise quadratic C^1 functions over $\mathcal{T}_i$ has dimension $3V_i + E_i$ with V_i, respectively E_i, the number of vertices, respectively, the edges of T_i. A function $g \in S_i$ will be uniquely defined by prescribing its function values and gradients at the vertices and the cross-derivatives at the midpoints of the edges of T_i. The interpolation procedure behind is completely local: the 12 degrees of freedom $f(P_j)$, $\nabla f(P_j)$, and the cross-derivatives $(\partial f/\partial n)(M_j)$, $j=1, 2, 3$, corresponding to some triangle $\Delta \in T_i$ (cf. Fig. 2) uniquely determine the piecewise quadratic function $I_i f$ inside Δ. The associated fundamental functions are considered in detail in [Ch-He] or in [Ch, Ex. 6.2].

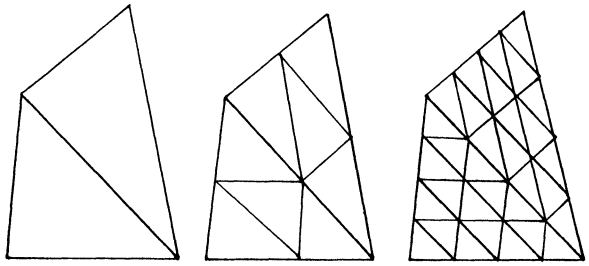


FIG. 1. *The dyadic refinement procedure.*

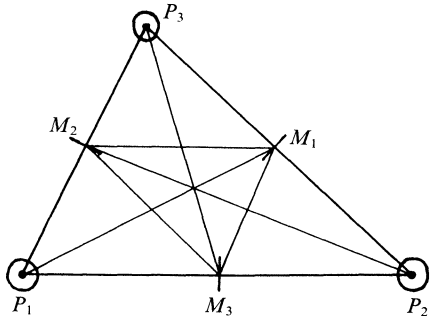


FIG. 2. *The Powell-Sabin piecewise quadratic C^1 element with 12 degrees of freedom (function value and first derivatives at P_j and cross-derivative at M_j, $j=1, 2, 3$).*

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To simplify further notation, let us introduce the following order in the set of interpolation points and conditions corresponding to T_i, $i=0, 1, \cdots$. For $i=0$ we number the vertices $P_{0,j}, j=1, \cdots, V_0$, and midpoints $M_{0,j}, j=1, \cdots, E_0$, in an arbitrary way. To each $P_{0,j}$ we associate two directions $n^1_{0,j}$ and $n^2_{0,j} \perp n^1_{0,j}$, to each $M_{0,j}$ the direction $n^3_{0,j}$ orthogonal to the edge (let $n^4_{0,j} \perp n^3_{0,j}$). For $i \geqq 1$, suppose that the sets $\{(P, n^1, n^2)_{i-1,j}\}\ j=1, \cdots, V_{i-1}$, and $\{(M, n^3)_{i-1,j}\}, j=1, \cdots, E_{i-1}$, have already been defined. Then put

$$(P, n^1, n^2)_{i,j} = \begin{cases} (P, n^1, n^2)_{i-1,j}, & j=1, \cdots, V_{i-1}, \\ (M, n^3, n^4)_{i-1,j-V_{i-1}}, & j=V_{i-1}+1, \cdots, V_i \end{cases}$$

(notice that $V_i - V_{i-1} = E_{i-1}$) and denote by $M_{i,j}$, $j=1, \cdots, E_i$, the midpoints of the edges in T_i (arbitrarily ordered) with $n^3_{i,j}$ the direction normal to the corresponding edge (once again, $n^4_{i,j} \perp n^3_{i,j}$).

Next we introduce two groups of functionals on $C^1(\bar{\Omega})$:

$$\varphi^0_{i,j}(f) = f(P_{i,j}), \qquad \varphi^1_{i,j}(f) = \frac{\partial f(P_{i,j})}{\partial n^1_{i,j}}, \quad j=1, \cdots, V_i,$$

$$\varphi^1_{i,j}(f) = \begin{cases} \dfrac{\partial f(P_{i,j-V_i})}{\partial n^2_{i,j-V_i}}, & j=V_i+1, \cdots, 2V_i, \\ \dfrac{\partial f(M_{i,j-2V_i})}{\partial n^3_{i,j-2V_i}}, & j=2V_i+1, \cdots, 2V_i+E_i. \end{cases}$$

The nodal basis

$$\mathcal{N}_i = \{N^0_{i,j}, j=1, \cdots, V_i\} \cup \{N^1_{i,j}, j=1, \cdots, 2V_i+E_i\}, \qquad i=0, 1, \cdots \tag{1.5}$$

of S_i will be defined by the following interpolation problems: find functions $N^0_{i,j} \in S_i$ such that

$$\begin{aligned} &\varphi^0_{i,s}(N^0_{i,j}) = 2^{-i}\delta_{j,s}, \qquad s=1, \cdots, V_i, \\ &\varphi^1_{i,s}(N^0_{i,j}) = 0, \qquad s=1, \cdots, 2V_i+E_i, \end{aligned} \tag{1.6}$$

and $N^1_{i,j} \in S_i$ such that

$$\begin{aligned} &\varphi^0_{i,s}(N^1_{i,j}) = 0, \qquad s=1, \cdots, V_i, \\ &\varphi^1_{i,s}(N^1_{i,j}) = \delta_{j,s}, \qquad s=1, \cdots, 2V_i+E_i. \end{aligned} \tag{1.7}$$

Consequently, the interpolation projection I_i associated with $\mathcal{N}_i$ is given by the formulae

$$I_i f = \sum_{j=1}^{V_i} 2^i \cdot \varphi^0_{i,j}(f) \cdot N^0_{i,j} + \sum_{j=1}^{2V_i+E_i} \varphi^1_{i,j}(f) \cdot N^1_{i,j}, \qquad f \in C^1(\Omega). \tag{1.8}$$

Besides the nodal basis (1.5) we introduce the so-called hierarchical basis $\mathcal{H}_i$ of S_i by

$$\mathcal{H}_i = \tilde{\mathcal{N}}_0 \cup \tilde{\mathcal{N}}_1 \cup \cdots \cup \tilde{\mathcal{N}}_i \qquad (\tilde{\mathcal{N}}_0 = \mathcal{N}_0), \tag{1.9}$$

where

$$\tilde{\mathcal{N}}_\nu = \{N^0_{\nu,j}, j=V_{\nu-1}+1, \cdots, V_\nu\} \cup \{N^1_{\nu,j}, j=V_\nu+V_{\nu-1}+1, \cdots, 2V_\nu+E_\nu\}$$

is the collection of all those nodal basis functions from $\mathcal{N}_\nu$ corresponding to degrees of freedom on T_ν not yet used on $T_{\nu-1}$ $(\nu=1, 2, \cdots)$.

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Let us list some obvious consequences of (1.2), (1.3), and the above construction without detailed proof:

$$\text{supp } N_{i,j}^k = \cup\{\bar{\Delta}: Q_{i,j}^k \in \bar{\Delta}\}, \qquad k = 0, 1, \tag{1.10}$$

where $Q_{i,j}^k$ is the interpolation point corresponding to $N_{i,j}^k$. Moreover, the number of different functions $N_{i,j}^k$ that do not vanish identically on a given $\Delta \in T_i$ does not exceed 12.

$$\|D^l N_{i,j}^k\|_{L_\infty} \leqq C_0 \cdot 2^{(l-1)i}, \qquad l = 0, 1, 2, \quad k = 0, 1, \tag{1.11}$$

where $C_0 > 0$ is an absolute constant only depending on T_0, and

$$D^l f = \sum_{m=0}^{l} \left| \frac{\partial^m f}{\partial x^m \partial y^{l-m}} \right|.$$

For all $g_i \in S_i$ and $1 \leqq p \leqq \infty$, we have

$$\|g_i\|_{L_p}^p \approx 2^{-2i} \cdot \left(\sum_{j=1}^{V_i} |\varphi_{i,j}^0(g_i)|^p + 2^{-ip} \cdot \sum_{j=1}^{2V_i+E_i} |\varphi_{i,j}^1(g_i)|^p \right) \tag{1.12}$$

(with an obvious modification if $p = \infty$).

Concerning the interpolation projection I_i, we have the important properties

$$I_i p = p \quad \text{on } \Delta \in T_i, \quad \text{whenever } p \text{ coincides with a quadratic polynomial on } \bar{\Delta}, \tag{1.13}$$

$$I_i g_\nu = g_\nu \quad \text{for all } g_\nu \in S_\nu, \quad \nu \leqq i. \tag{1.14}$$

The L_p-approximation theory of the subspaces S_i, which is mainly based on the above properties, will be developed in § 2 below.

Remark 1.1. Other quasi interpolants on S_i preserving (1.13) may be introduced as well. For example, in [Ch-He] an interpolation projection

$$\tilde{I}_i f = \sum_{k=0}^{1} \sum_{j} \tilde{\varphi}_{i,j}^k(f) \cdot N_{i,j}^k, \qquad f \in C^1(\bar{\Omega}),$$

where the functionals $\tilde{\varphi}_{i,j}^k$ involve only linear combinations of function values $f(P_{i,s})$ is described for type-I triangulations.

1.2. Clough–Tocher type elements. The use of the original piecewise cubic Clough-Tocher element (in the same spirit as above) leads to an obvious difficulty: The sequence of induced triangulations $\{\mathcal{T}_i\}$ will not be monotone, which implies the undesired property $S_i \not\subseteq S_{i+1}$ where $S_i \equiv S_3^1(\mathcal{T}_i)$ (cf. Fig. 3).

We therefore introduce a slightly modified Clough-Tocher type element that avoids this difficulty (see Fig. 4). Formally, it looks like the Powell-Sabin element but carries

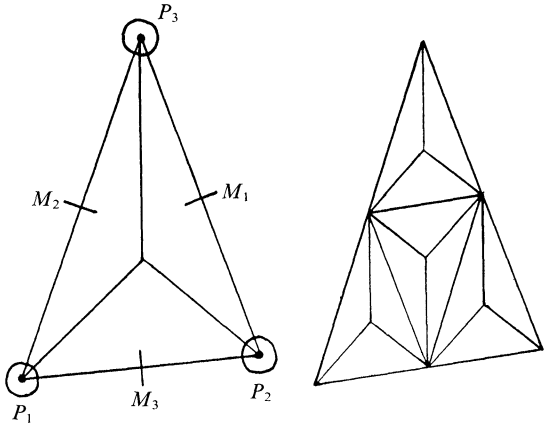


FIG. 3. *The Clough-Tocher element, and the illustration of* $\mathcal{T}_i \not\prec \mathcal{T}_{i+1}$.

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more degrees of freedom in order to define the piecewise cubic C^1 function in the macroelement in a unique way. Due to Schumaker's dimension result [Sch, Thm. 2.2], the dimension of the macroelement is 30, with a possible choice of these degrees of freedom as proposed in Fig. 4. A brief look at the Bezier net of the macroelement shows the correctness of this proposal (cf. [Fa] for the terminology and the understanding of Fig. 5).

As in § 1.1 we introduce now the increasing sequence $\{\mathcal{T}_i\}$, the finite element subspaces $S_i \equiv S_3^1(\mathcal{T}_i)$ of piecewise cubic C^1 functions on $\mathcal{T}_i$, equipped with the analogously defined basis systems $\mathcal{N}_i$ and $\mathcal{H}_i$ (once again, nodal basis functions $N_{i,j}^0$ corresponding to function evaluations are normalized by a factor 2^{-i}). Properties (1.10)–(1.14) carry over to the new situation, with some obvious changes, e.g., the projection I_i now preserves cubic polynomials:

(1.13)′ $\quad I_i p = p \quad$ in $\Delta \in T_i \quad$ whenever p is a cubic polynomial on $\bar{\Delta}$.

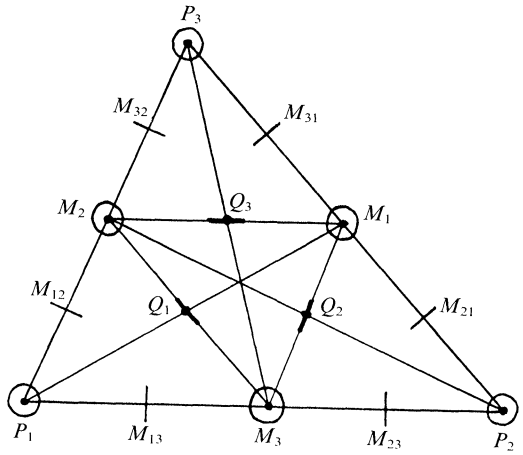


FIG. 4. *The Clough–Tocher type piecewise cubic C^1 element with 30 degrees of freedom (function value and first derivatives at P_i, M_i, cross-derivative at M_{ij}, function value and one directional derivative at Q_i).*

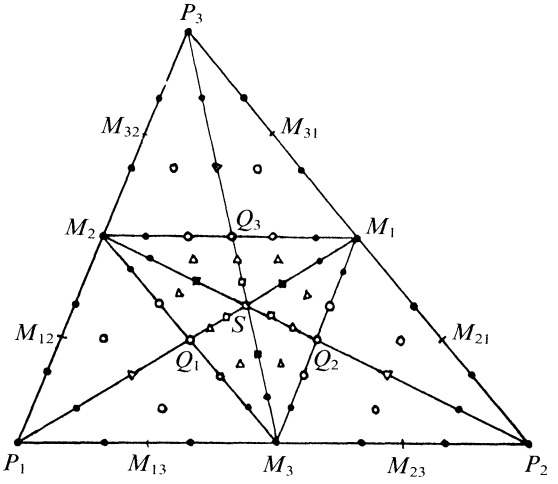


FIG. 5. *The B-net of the Clough–Tocher type macroelement.*

● *from the data at P_i, M_i* △ *C^1 condition along M_iM_j*
○ *from the data at Q_i, M_{ij}* ■ *C^1 condition along SM_i*
▽ *C^1 condition along Q_iP_i* □ *C^1 condition at S*

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Remark 1.2. It is natural to compare the complexity of the above construction on some fixed T_i with that of the Powell-Sabin construction on T_{i+1}. Although in doing so we easily observe that the number of degrees of freedom is only slightly increased (by about 10%), note that the basis functions are less local, which results in less sparsity of the nodal (and hierarchical) discretizations of (0.2). The advantage is that a higher degree of approximation is available provided that the solution of (0.1) is smooth enough ($u \in H^4(\Omega)$).

Remark 1.3. In both cases, when solving (0.1)-(0.2) the construction must be adapted to the essential boundary conditions, i.e., we must exclude all those basis functions from $\mathcal{N}_i$, respectively, $\mathcal{H}_i$, that correspond to boundary nodes, and consider $S_{i,0} = S_d^1(\mathcal{T}_i) \cap H_0^2(\Omega)$ instead of S_i.

2. Approximation spaces related to C^1 finite elements. We follow the strategy of [Os1]: Starting with a sequence of suitable ("spline-like and dyadically refined") subspaces $S_0 \subseteq S_1 \subseteq \cdots$ we define the scale of $A_{p,q}^s(\{S_i\})$-spaces as follows: A function $f \in L_p(\Omega)$ belongs to $A_{p,q}^s \equiv A_{p,q}^s(\{S_i\})$ for some fixed $s \geqq 0$, $1 \leqq p, q \leqq \infty$ if

(2.1) There exists a sequence $g_i \in S_i$, $i = 0, 1, \cdots$ such that $f = \sum_{i=0}^{\infty} g_i$ in the sense of L_p, and $\|\{2^{is} \cdot \|g_i\|_{L_p}\}\|_{l_q} < \infty$.

Moreover,

$$\|f\|_{A_{p,q}^s} = \inf \|\{2^{is} \cdot \|g_i\|_{L_p}\}\|_{l_q} \tag{2.2}$$

defines a norm on $A_{p,q}^s$ (the infimum in (2.2) must be taken with respect to all admissible representations (2.1) of f). If $s = 0$, only $q = 1$ makes sense.

Constructions of this type have a long history; see [N], [Be-I-N], and [T] for their trigonometric counterparts (involving decompositions of the Fourier transform). Also spline and wavelet constructions have been widely used (see, e.g., [D-Pop2], [Si], and [Me]). The interest in finite element based spaces is motivated by the obvious close relationship between the above definition and some actual problems in the analysis of the finite element method (such as multigrid-multilevel schemes, optimal grid generation by adaptive methods, etc.). Note that the most important particular case is $p = q = 2$; applications of the full range of parameters $0 < p, q \leqq \infty$ can be found in [Os1] where the case of C^0 Lagrange elements is studied.

From now on, let $\{S_i\}$ be one of the sequences of C^1 finite element subspaces introduced in the previous section; cf. also Remark 1.3 for the modifications in the case of essential boundary conditions. The first assertion is standard; we refer to [Os1, § 1] for details and further references.

PROPOSITION 1. *Let $A_{p,q}^s \equiv A_{p,q}^s(\{S_i\})$ be defined by* (2.1), (2.2) *where $\{S_i\}$ is defined in* §§ 1.1-1.2. *Then the spaces $A_{p,q}^s$ are Banach spaces for all parameter choices indicated above, and*

$$A_{p,1}^0 \cong L_p, \qquad 1 \leqq p < \infty, \tag{2.3}$$

$$\|f\|_{A_{p,q}^s}^{(1)} \equiv \|f\|_{L_p} + \|\{2^{is} \cdot E_i(f)_p\}\|_{l_q}, \qquad s > 0, \tag{2.4}$$

is an equivalent norm where

$$E_i(f)_p = \inf_{g_i \in S_i} \|f - g_i\|_{L_p}$$

are the best approximations of f with respect to S_i.

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$$\|f\|^{(2)}_{A^s_{p,q}} \equiv \|R_0 f\|_{L_p} + \|\{2^{is}\cdot \|R_i f - R_{i-1} f\|_{L_p}\}\|_{l_q}, \qquad s>0, \tag{2.5}$$

is an equivalent norm whenever the linear mappings $R_i : L_p \to S_i$ *satisfy (uniformly in* i*)*

$$\|f - R_i f\|_{L_p} \leqq C \cdot E_i(f)_p.$$

Remark 2.1. Norms of type (2.5) can also be found in [D-Pop2]. Other practically relevant equivalent norms can be given by using (1.12). Embedding theorems of Sobolev type are also a trivial consequence of the definition of the spaces and (1.12). In the case $p = q = 2$, we derive from (2.4) (or (2.5)) that

$$\|f\|_{A^s_{2,2}} \approx \|Q_0 f\|_{L_2} + \|\{2^{is}\cdot \|Q_i f - Q_{i-1} f\|_{L_2}\}\|_{l_2}, \qquad s>0, \tag{2.6}$$

where $Q_i : L_2 \to S_i$ are the canonical orthoprojections, $i = 0, 1, \cdots$. In § 3, in connection with the condition number estimates, we will study norms of the same type with Q_i replaced by I_i. For a direct application of (2.6), see Remark 3.4.

In order to work with the abstract $A^s_{p,q}$-spaces in applications, we must relate them to the more convenient function spaces of Besov–Sobolev type. Let $B^s_{p,q} \equiv B^s_{p,q}(\Omega)$, $s>0$, $1 \leqq p, q \leqq \infty$, denote the Besov spaces, i.e., the Banach spaces of functions $f \in L_p$ with the norm

$$\|f\|_{B^s_{p,q}} \equiv \|f\|_{L_p} + \|\{2^{is}\cdot \omega_m(2^{-i}, f)_p\}\|_{l_q} < \infty,$$

where

$$\omega_m(t,f)_p = \sup_{|h|\leqq t} \|\Delta^m_h f\|_{L_p}, \qquad t>0, \quad m = 1, 2, \cdots \tag{2.7}$$

denotes the (total) L_p-modulus of continuity of order m of f. The choice of the integer $m > s$ in the definition of the Besov spaces is arbitrary (up to equivalent norms). It is well known that

$$H^s(\Omega) \cong W^s_2(\Omega) \cong B^s_{2,2}(\Omega), \qquad s>0, \tag{2.8}$$

with $H^s \equiv H^s(\Omega)$, respectively, $W^s_p \equiv W^s_p(\Omega)$, the Sobolev–Slobodetzkij spaces (L_2, respectively, L_p case, $1 \leqq p \leqq \infty$), $s>0$, cf. [Be-I-N] or [T]. For arbitrary $p \geqq 1$ we have

$$B^s_{p,\min(p,2)} \subset W^s_p \subset B^s_{p,\max(p,2)}, \qquad s>0. \tag{2.9}$$

PROPOSITION 2. *For* $1 \leqq p, q \leqq \infty$, $s>0$, *and the spaces introduced above we have*

$$B^s_{p,q} \subset A^s_{p,q}, \qquad 0<s<d+1, \tag{2.10}$$

where d *is the polynomial degree of the elements under consideration* ($d=2$ *for the Powell–Sabin elements,* $d=3$ *for the Clough–Toucher type elements*).

Conversely,

$$A^s_{p,q} \subset B^s_{p,q}, \qquad 0<s<2+\frac{1}{p}, \tag{2.11}$$

and thus,

$$A^s_{p,q} \cong B^s_{p,q}, \qquad 0<s<2+\frac{1}{p}. \tag{2.12}$$

Proof. We will apply the standard technique of Jackson–Bernstein inequalities. First, let us verify the Jackson type estimate

$$E_i(f)_p \leqq C \cdot \omega_{d+1}(2^{-i}, f)_p, \tag{2.13}$$

where the constant $C>0$ is independent of $i=0,1,\cdots$ and $f\in L_p$. According to the characterization of moduli of continuity by K'-functionals (cf. [Da-D-S] for the result and some references),

$$\omega_m(t,f)_p \approx \inf_{g\in W_p^m} (\|f-g\|_{L_p} + t^m\cdot\|D^m g\|_{L_p}), \qquad t\to 0; \tag{2.14}$$

the inequality (2.13) will immediately follow from some particular result for functions from W_p^{d+1}:

$$\|g-I_i g\|_{L_p} \leqq C\cdot 2^{-i(d+1)}\cdot\|D^{d+1}g\|_{L_p}, \qquad g\in W_p^{d+1}. \tag{2.15}$$

(Indeed, if (2.15) is true, then

$$\begin{aligned}\|f-I_i g\|_{L_p} &\leqq \|f-g\|_{L_p} + \|g-I_i g\|_{L_p}\\ &\leqq C\cdot(\|f-g\|_{L_p} + 2^{-i(d+1)}\|D^{d+1}g\|_{L_p}),\end{aligned}$$

and taking the infimum with respect to $g\in W_p^{d+1}$ we get (2.13).) To prove (2.15), suppose that $1+2/p<d+1$. Then, by the Sobolev embeddings we have $W_p^{d+1}\subset C^1(\Omega)$ and thus, $I_i g$ is well defined for any $g\in W_p^{d+1}$. Since for any triangle $\Delta\in T_i$, and any polynomial q of degree d,

$$\|g-I_i g\|_{L_p(\Delta)} \leqq \|g-q\|_{L_p(\Delta)} + \|I_i(g-q)\|_{L_p(\Delta)}$$

(cf. (1.13), respectively, (1.13)'), we can now make use of the locality of the projection (see also (1.11), (1.12), and their counterpart for the other element type)

$$\|I_i(g-q)\|_{L_p(\Delta)}^p \leqq C\cdot 2^{-2i}\sum_{k=0}^{1} 2^{kpi}\cdot\|D^k(g-q)\|_{C(\Delta)}^p.$$

Substituting here the Sobolev inequalities

$$\|D^k(g-q)\|_{C(\Delta)} \leqq C\cdot 2^{i(2/p-k)}(\|g-q\|_{L_p(\Delta)} + 2^{-i(d+1)}\|D^{d+1}g\|_{L_p(\Delta)}),$$

where $2/p+1<d+1$, $k=0$, 1, into the latter expression, we obtain

$$\begin{aligned}\|g-I_i g\|_{L_p(\Delta)}^p &\leqq C\cdot\inf_q(\|g-p\|_{L_p(\Delta)}^p + 2^{-ip(d+1)}\|D^{d+1}g\|_{L_p(\Delta)}^p)\\ &\leqq C\cdot 2^{-ip(d+1)}\cdot\|D^{d+1}g\|_{L_p(\Delta)}^p,\end{aligned}$$

where in the last step the Whitney–Jackson inequality for algebraic polynomials was used (see once again [Da-D-S]). Thus, after summing up with respect to all triangles in T_i we establish (2.15) under the restriction $1+2/p<d+1$. The latter condition is fulfilled except one case ($p=1$ for the Powell–Sabin elements where (2.15) does not make sense since $W_1^3\not\subset C^1$). However, (2.13) remains valid in this case, too. To this end, it suffices to "correct" the definition of I_i by replacing all first derivatives by suitable differences of function values such that (1.13) remains unchanged (cf. Remark 1.1). With this new projection at hand we immediately get an analogue of (2.15) also for $p=1$ which is sufficient for the proof of (2.13) in the exceptional case. Since we will not rely on this result in the further considerations, the details are left to the reader.

The imbedding (2.10), together with the corresponding norm estimate, is now obvious from (2.15) and (2.4). To get (2.11) as well, we need the so-called Bernstein type (or inverse) inequality

$$\omega_{d+1}(t,g_i)_p \leqq C\cdot(\min\{1,2^i t\})^{2+1/p}\cdot\|g_i\|_{L_p}, \qquad g_i\in S_i. \tag{2.16}$$

Since for $t\geqq 2^{-i}$ this inequality reduces to

$$\omega_{d+1}(t,f)_p \leqq C\cdot\|g_i\|_{L_p},$$

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which directly follows from the definition (2.17) with $C=2^{d+1}$, we concentrate on $t<2^{-i}$. First, according to the representation

$$g_i = I_i g_i = \sum_{k=0}^{1} 2^{(1-k)i} \sum_j \varphi_{i,j}^k(g_i) \cdot N_{i,j}^k,$$

we get

$$\Delta_h^{d+1} g_i = \sum_{k=0}^{1} 2^{(1-k)i} \sum_j \varphi_{i,j}^k(g_i) \cdot \Delta_h^{d+1} N_{i,j}^k.$$

Observe that $\operatorname{mes}(\operatorname{supp} \Delta_h^{d+1} N_{i,j}^k) \leqq C \cdot 2^{-i} t$ for all $|h| \leqq t$ because $N_{i,j}^k$ is a piecewise polynomial of degree d with respect to $\mathcal{T}_i$ and vanishes outside a small part of Ω. Therefore, relying on (1.10), (1.11), (1.12) (and their counterparts), we get for $\|h\| \leqq t < 2^{-i}$

$$\begin{aligned}
\|\Delta_h^{d+1} g_i\|_{L_p}^p &\leqq C \cdot \sum_{k=0}^{1} 2^{(1-k)pi} \sum_j |\varphi_{i,j}^k(g_i)|^p \cdot \|\Delta_h^{d+1} N_{i,j}^k\|_{L_p}^p \\
&\leqq C \cdot \sum_{k=0}^{1} 2^{(1-k)pi} \sum_j |\varphi_{i,j}^k(g_i)|^p \cdot \operatorname{mes}(\operatorname{supp} \Delta_h^{d+1} N_{i,j}^k) \cdot (|h^2| \cdot \|D^2 N_{i,j}^k\|_{L_\infty})^p \\
&\leqq C \cdot \sum_{k=0}^{1} 2^{(1-k)pi} \sum_j |\varphi_{i,j}^k(g_i)|^p \cdot t^{2p+1} \cdot 2^{(p-1)i} \\
&= C \cdot (2^i t)^{2p+1} \cdot \sum_{k=0}^{1} 2^{-(2+kp)i} \sum_j |\varphi_{i,j}^k(g_i)|^p \\
&\leqq C \cdot ((2^i t)^{2+1/p} \cdot \|g_i\|_{L_p})^p,
\end{aligned}$$

which yields (2.16).

Though the derivation of (2.11) from (2.16) is standard, we give some details for the sake of completeness. Let $f \in A_{p,q}^s$, and let $f = \sum_{i=0}^{\infty} g_i$, $g_i \in S_i$, be an admissible representation (see (2.1)). Then

$$\|f\|_{L_p} \leqq \sum_{i=0}^{\infty} \|g_i\|_{L_p}$$

and (with $\lambda = 2 + 1/p$)

$$\omega_{d+1}(2^{-j}, f)_p \leqq \sum_{i=0}^{\infty} \omega_{d+1}(2^{-j}, g_i)_p \leqq C \left(\sum_{i=0}^{j} 2^{(i-j)\lambda} \|g_i\|_{L_p} + \sum_{i=j+1}^{\infty} \|g_i\|_{L_p} \right).$$

Thus,

$$\begin{aligned}
\|f\|_{B_{p,q}^s} &\leqq C \cdot \left[\left\| \left\{ 2^{j(s-\lambda)} \sum_{i=0}^{j} 2^{i\lambda} \|g_i\|_{L_p} \right\} \right\|_{l_q} + \left\| \left\{ 2^{js} \sum_{i=j+1}^{\infty} \|g_i\|_{L_p} \right\} \right\|_{l_q} \right] \\
&\leqq C \cdot \|\{2^{is} \|g_i\|_{L_p}\}\|_{lq}, \qquad 0 < s < \lambda,
\end{aligned}$$

where the last step is based on the well-known Hardy type inequalities for sequences. Now, taking the infimum with respect to all admissible representations (2.1) of f, we finally obtain (2.11).

Remark 2.2. Note that the trace problem for the above approximation spaces can be studied in full analogy to [Os1, § 1] where C^0 element based $A^s_{p,q}$-spaces were considered in detail. As a consequence, if we replace S_i by $S_{i,0}$ (cf. Remark 1.3) then the definitions (2.1)-(2.2) lead to equivalent norms in $B^s_{p,q,0}(\Omega)$, the closure of $C_0^\infty(\Omega)$ in the $B^s_{p,q}$-norm, if $1+1/p<s<2+1/p$. In particular,

$$\|f\|^2_{H^2} \approx \inf \sum_{i=0}^{\infty} 2^{4i} \cdot \|g_i\|^2_{L_2}, \quad f=\sum_{i=0}^{\infty} g_i, \quad g_i \in S_{i,0}, \quad i \geqq 0, \tag{2.17}$$

for any $f \in H^2_0(\Omega)$.

3. Hierarchical norm estimates and condition numbers. In this section we consider some norm inequalities for finite element functions introduced in § 1, and derive the growth estimates for the condition numbers of the matrices corresponding to the discretizations of the biharmonic problem (0.1)-(0.2).

Let $\bar{g}_j \in S_j$, $j=1,2,\cdots$, be arbitrary but fixed. By definition, $\bar{g}_j \in A^s_{p,q}(\{S_i\})$ for all reasonable parameters. The $A^s_{p,q}$-norms of $\bar{g}_j$ can be checked by the following simplified formulae (which are exact up to absolute constants):

$$\|\bar{g}_j\|_{A^s_{p,q}} \approx \inf \|\{2^{is} \cdot \|g_i\|_{L_p}\}_{i=0,\cdots,j}\|_{l_q}, \qquad \bar{g}_j = \sum_{i=0}^{j} g_i, \tag{3.1}$$

where the infimum is taken with respect to all representations (2.1) of $f=\bar{g}_j$ with $g_i \equiv 0$ for $i>j$ (this is an obvious consequence of (2.4)).

In order to simplify the matter, we restrict our attention from now on to the most relevant case $p=q=2$. From Proposition 2 and (2.8) we get

$$A^s_{2,2} \cong H^s, \quad 0<s<\tfrac{5}{2}, \qquad H^s \subset A^s_{2,2}, \quad \tfrac{5}{2} \leqq s \leqq d+1. \tag{3.2}$$

The first task is to compare $\|\bar{g}_j\|_{H^2} \approx \|\bar{g}_j\|_{A^2_{2,2}}$ given by (3.1) with the norm

$$\|\bar{g}_j\|_j^2 \equiv \|I_0 \bar{g}_j\|^2_{L_2} + \sum_{i=1}^{j} 2^{4i} \cdot \|I_i \bar{g}_j - I_{i-1}\bar{g}_j\|^2_{L_2} \tag{3.3}$$

(which differs slightly from the discrete norm (0.3) mentioned in the Introduction).

Obviously, since $I_0\bar{g}_j \in S_0$, $I_i\bar{g}_j - I_{i-1}\bar{g}_j \in S_i$, $i=1,\cdots,j$, we have

$$C \cdot \|\bar{g}_j\|_{H^2} \leqq \|\bar{g}_j\|_{A^2_{2,2}} \leqq \|\bar{g}_j\|_j,$$

with some $C>0$ (cf. (3.1)). To derive the opposite inequality, let $\bar{g}_j$ be arbitrarily decomposed as given in (3.1). Then

$$\|\bar{g}_j\|_j^2 = \left\| g_0 + I_0\left(\sum_{m=1}^{j} g_m\right)\right\|^2_{L_2} + \sum_{i=1}^{j} 2^{4i} \left\| g_i + I_i\left(\sum_{m=i+1}^{j} g_m\right) - I_{i-1}\left(\sum_{m=i}^{j} g_m\right)\right\|^2_{L_2}$$

$$\leqq C \cdot \left(\|\{2^{2i}\|g_i\|_{L_2}\}\|^2_{l_2} + \left\| \left\{ 2^{2i} \left\| I_i\left(\sum_{m=i+1}^{j} g_m\right)\right\|_{L_2}\right\}\right\|^2_{l_2}\right) \equiv C \cdot (A+B).$$

According to (1.12) and its analogue for the case of the Clough-Tocher type element, we get for the second summand

$$B \leqq C \cdot \sum_{i=0}^{j} 2^{4i} \left(2^{-2i} \sum_{k=0}^{1} 2^{-2ki} \sum_{s} \left(\sum_{m=i+1}^{j} \varphi^k_{i,s}(g_m)\right)^2\right)$$

(note that the summation over s has to be carried out, for each k, with respect to all functionals $\varphi^k_{i,s}$ corresponding to T_i).

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Now we estimate (with $\alpha_0 = 1$, $\alpha_1 = 0$)

$$\left(\sum_{m=i+1}^{j} \varphi_{i,s}^{k}(g_m)\right)^2 = \left(\sum_{m=i+1}^{j} 2^{-\alpha_k(m-i)} \cdot (2^{\alpha_k(m-i)} \cdot \varphi_{i,s}^{k}(g_m))\right)^2$$

$$\leqq \left(\sum_{r=1}^{j-i} 2^{-\alpha_k r}\right) \cdot \sum_{m=i+1}^{j} 2^{\alpha_k(m-i)} \cdot \varphi_{i,s}^{k}(g_m)^2$$

$$\leqq C \cdot \begin{cases} \sum_{m=i+1}^{j} 2^{m-i} \cdot \varphi_{i,s}^{0}(g_m)^2, & k=0, \\ (j-i) \cdot \sum_{m=i+1}^{j} \varphi_{i,s}^{1}(g_m)^2, & k=1, \end{cases}$$

and substitute into the previous estimate for B. Changing the order of summation, this yields

$$B \leqq C \cdot \sum_{m=1}^{j} 2^{4m} \left\{ 2^{-3m} \cdot \sum_{i=0}^{m-1} 2^i \cdot \sum_{s'} \varphi_{m,s'}^{0}(g_m)^2 + 2^{-4m} \cdot \sum_{i=0}^{m-1} (j-i) \cdot \sum_{s'} \varphi_{m,s'}^{1}(g_m)^2 \right\}$$

$$\leqq C \cdot j^2 \sum_{m=1}^{j} 2^{4m} \cdot \|g_m\|_{L_2}^2$$

(cf. once again (1.12), $\cdots$) and, finally,

$$\|\bar{g}_j\|_j^2 \leqq C \cdot j^2 \cdot \|\{2^{2i} \cdot \|g_i\|_{L_2}\}\|_{l_2}^2.$$

After taking the infimum with respect to all decompositions (3.1) of $\bar{g}_j$, we have the desired result.

PROPOSITION 3. *Let $\bar{g}_j \in S_j$, $j = 1, 2, \cdots$, be arbitrary where $\{S_i\}$ is one of the sequences of C^1 finite element subspaces constructed in § 1. Then*

$$C_1 \cdot \|\bar{g}_j\|_{H^2} \leqq \|\bar{g}_j\|_j \leqq C_2 \cdot j \cdot \|\bar{g}_j\|_{H^2}. \tag{3.4}$$

The constants C_1, C_2 are independent of $\bar{g}_j$ and the number of refinement levels j.

Note that according to Remark 2.2 the result of (3.4) remains true if $\{S_i\}$ is replaced by $\{S_{i,0}\}$. As a corollary we can now state the main result.

THEOREM. *Denote by A_H the matrix of the discretized variational problem*

$$\text{Find } \hat{g}_j \in S_{j,0} \text{ such that } a(\hat{g}_j, g_j) = (f, g_j) \text{ for all } g_j \in S_{j,0} \tag{3.5}$$

(cf. (0.1)-(0.2)) corresponding to the hierarchical basis $\mathcal{H}_{j,0}$ in $S_{j,0}$. Then the condition number $\kappa(A_H)$ satisfies

$$\kappa(A_H) \leqq C_3 \cdot j^2, \qquad j = 1, 2, \cdots, \tag{3.6}$$

for the Powell-Sabin and Clough-Tocher type constructions introduced above. For comparison, the nodal basis discretization matrix A_N corresponding to $\mathcal{N}_{j,0}$ satisfies $\kappa(A_N) \approx 2^{4j}$.

Proof. Let $\bar{g}_j \in S_{j,0}$ be represented by the hierarchical basis as

$$\bar{g}_j = \sum_{i=0}^{j} \sum_{k=0}^{1} \sum_{s \in \tilde{\mathcal{N}}_{i,0}} x_{i,s}^{k} \cdot N_{i,s}^{k},$$

where $s \in \tilde{\mathcal{N}}_{i,0}$ indicates that only those s corresponding to interior nodal basis functions in $\tilde{\mathcal{N}}_i$ come into play. By the definition of $\mathcal{H}_{j,0}$ the coefficients $x_{i,s}^{k}$ that we arrange to the vector

$$x_H = (\underbrace{\cdots, x_{0,s}^{k}}_{\tilde{\mathcal{N}}_{0,0}}, \underbrace{\cdots, x_{1,s}^{k}}_{\tilde{\mathcal{N}}_{1,0}}, \underbrace{\cdots, x_{j,s}^{k}, \cdots}_{\cdots \tilde{\mathcal{N}}_{j,0}})^T$$

(the order within the groups must coincide with the fixed order of the basis functions in $\tilde{\mathcal{N}}_i$; cf. (1.9) and Remark 1.3) are given by

$$x_{i,s}^k = 2^{(1-k)i} \cdot \varphi_{i,s}^k (I_i \bar{g}_j - I_{i-1} \bar{g}_j) \qquad (I_{-1} \bar{g}_j \equiv 0). \tag{3.7}$$

On the one hand, we have

$$((A_H x_H, x_H)) = a(\bar{g}_j, \bar{g}_j) \approx \|\bar{g}_j\|_{H^2}^2, \tag{3.8}$$

where $((\cdot, \cdot))$ denotes the corresponding scalar product for vectors. On the other hand, by (3.7) and (1.12)

$$\begin{aligned} ((x_H, x_H)) &= \sum_{i=0}^{j} \sum_{k=0}^{1} 2^{2(1-k)i} \sum_{s \in \tilde{\mathcal{N}}_{i,0}} \varphi_{i,s}^k (I_i \bar{g}_j - I_{i-1} \bar{g}_j)^2 \\ &\approx \sum_{i=0}^{j} 2^{4i} \cdot \|I_i \bar{g}_j - I_{i-1} \bar{g}_j\|_{L_2}^2 \approx \|\bar{g}_j\|_j^2 . \end{aligned} \tag{3.9}$$

The constants in the two-sided estimates stated in (3.8)–(3.9) are independent of $\bar{g}_j$ and $j = 1, 2, \cdots$. By Proposition 3 we arrive at

$$\bar{C}_2 \cdot j^{-2} \leqq \frac{((A_H x_H, x_H))}{((x_H, x_H))} \leqq \bar{C}_1 . \tag{3.10}$$

Since $\bar{g}_j$ and therefore x_H were arbitrary, (3.10) yields (3.6) according to the definition of the spectral condition numbers $\kappa(A)$ for positive definite matrices.

In the case of A_N, an analogous consideration results in

$$\frac{((A_N x_N, x_N))}{((x_N, x_N))} \approx \frac{2^{-4j} \cdot \|\bar{g}_j\|_{H^2}}{\|\bar{g}_j\|_{L_2}} . \tag{3.11}$$

Using (3.2), (3.1), we immediately obtain the upper bound

$$\frac{((A_N x_N, x_N))}{((x_N, x_N))} \leqq C$$

(put $g_i = 0$ for $i < j$, $g_j = \bar{g}_j$, and $p = q = 2$ in (3.1)). On the other hand, $\|u\|_{H^2} \geqq C \cdot \|u\|_{L_2}$ for arbitrary $u \in H_0^2$. Thus, $\kappa(A_N) \leqq C \cdot 2^{4j}$ is established. To get the sharpness of this upper bound, we must consider the following two examples: $\bar{g}_j = N_{j,s}^0$, and $\bar{g}_j = N_{1,s}^0$ for some s. The details are left to the reader.

Remark 3.1. It can be shown that the estimates (3.4), respectively (3.6), are asymptotically sharp with respect to the number j of refinement levels. The corresponding examples can be constructed by adapting the considerations in [Y1], respectively [Os2], to the new cases. For example, fixing some interior vertex P^* of, say, T_2 and considering the basis functions $N_i^* = N_{i,j}^1$, $i \geqq 2$, corresponding to P^* in $S_{i,0}$, we can construct examples

$$g_j^* = \sum_{i=2}^{j} N_i^* \in S_{j,0}, \qquad j \geqq 2,$$

which show the sharpness of the nontrivial upper estimate in (3.4), respectively the lower estimate in (3.10). The details are left to the reader.

Remark 3.2. The practical solution of (3.5) by the conjugate gradient method applied to the hierarchical discretization can be organized as proposed for the second-order Poisson equation in [Y1]. Let $\bar{g}_j \in S_j$; as above we denote by x_H (respectively, x_N) the coefficient vector of $\bar{g}_j$ with respect to the hierarchical (respectively, the nodal

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basis). Let L be the transfer matrix between the two bases, i.e., $x_N = Lx_H$. Then we easily derive that

$$A_H = L^T A_N L.$$

This representation suggests the use of $C_H = L \cdot L^T$ as preconditioner for the nodal discretization of (3.5). Note that as in the case of [Y1] the multiplication by L, respectively, L^T, can be decomposed into a sequence of elementary matrix multiplications

$$L = L_j \cdot \cdots \cdot L_2 \cdot L_1 \qquad \text{(respectively, } L^T = L_1^T \cdot L_2^T \cdot \cdots \cdot L_j^T\text{)},$$

where the matrices L_i correspond to a transfer step involving the two grids T_{i-1} and T_i, only. The necessary formulae can easily be derived from (3.7); we observe that the multiplication by L_i can be carried out "locally," i.e., recomputing a few coefficients corresponding to each triangle (respectively, to the edges of T_{i-1}). Moreover, the multiplication by L_i requires $O(2^{2i})$ flops which results in $O(2^{2j})$ arithmetical operations for the complete transfer step (the multiplication by L^T satisfies analogous properties).

Neglecting for a while the considerable overhead time, the preconditioned conjugate gradient method just described will lead to an almost optimal algorithm: It needs $O(j \cdot N \cdot |\log(\varepsilon/2)|)$ flops to reduce the initial error by a factor ε. Here $N \approx h(T_j)^{-2} \approx 2^{2j}$ denotes the dimension of the linear system corresponding to the nodal basis discretization. This means that, asymptotically, the method is only slightly more complex (by a factor $\approx \log N$) than multigrid schemes with conforming or nonconforming elements which usually require $O(N \cdot |\log(\varepsilon/2)|)$ operations (see [P-Brae], [Br]).

However, the restriction to conforming elements generally leads to more computations per iteration step, i.e., both the coefficient matrix A_N and the preconditioner $C = L \cdot L^T$ are more dense. A verification of the computational performance of the above hierarchical multilevel method for the biharmonic equation is still in progress, and will be reported on elsewhere. A prospectively more interesting problem is to develop the hierarchical approach if one of the basic assumptions of the above theory, i.e., that $S_0 \subseteq \cdots \subseteq S_j \subseteq \cdots$, is violated (this is just the case for nonconforming methods, for the original Clough–Tocher elements as discussed in § 1.2, or the classical Argyris triangles). For some results in this direction, see [Dö1], [Dö2], and [Os5].

Remark 3.3. An important feature of the hierarchical method is that local refinement and adaptivity can be incorporated in an uncomplicated manner. There are two slightly different strategies in that respect. The usual one followed by most of the authors restricts the flexibility of grid generation to a combination of regular dyadic refinements (as described at the beginning of § 1) with some irregularly refined triangles which, however, may not be further refined. Such restrictions are acceptable from a practical viewpoint; see [B] and [De-L-Y]. Since under these circumstances triangles cannot degenerate, and due to the local character of all constructions, the above theory can be extended without substantial changes (cf. also [Y1], [Y4], and [Os2] in the case of C^0 elements and second-order elliptic problems).

Since for composite C^1 elements (as well as for other more complex element types) the structure of the irregular elements is complicated, an alternative approach might be useful: instead of locally refining the underlying triangulation, we may select only part of the basis functions from $\tilde{\mathcal{N}}_{i,0}$ to form the next finite element subspace $S_{i,0}^*$. More precisely, let

$$S_{i,0}^* = \operatorname{span}\{N_{m,s}^k : N_{m,s}^k \in \mathcal{H}_{i,0}^* (m \leqq i)\},$$

where $\mathcal{H}_{i,0}^* = \mathcal{N}_{0,0} \cup \tilde{\mathcal{N}}_{1,0}^* \cup \cdots \cup \tilde{\mathcal{N}}_{i,0}^*$ with $\tilde{\mathcal{N}}_{i,0}^* \subseteq \tilde{\mathcal{N}}_{i,0}$, $i = 0, \cdots, j$, is a certain subset of $\mathcal{H}_{i,0}$. This construction automatically assures that $\kappa(A_H^*) \leqq \kappa(A_H)$. Also, A_N^* and the

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transfer steps, i.e., the matrices L^* and $(L^*)^T$, can easily be derived from the case of uniform refinements briefly discussed in Remark 3.2. Note that there are some theoretical results [Os4], [Os1], [D-Pop1] on how to select basis functions in order to achieve asymptotically optimal approximation, respectively, convergence rates (in the sense of the h-version of the finite element method). Clearly, the two approaches are closely related.

Remark 3.4. Our analysis also allows us to design a preconditioner analogous to that proposed by Bramble, Pasciak, and Xu [B-P-X, X] for second-order problems. For example, consider the Powell–Sabin case and let

$$\bar{g}_j = \sum_{k=0}^{1} \sum_{s} x_{s,k} N_{j,s}^k \in S_{j,0} \qquad (x_N = \{x_{s,k}\})$$

be the nodal basis representation of $\bar{g}_j$. Using the splitting

$$\bar{g}_j = Q_0 \bar{g}_j + \sum_{i=0}^{j} (Q_i \bar{g}_j - Q_{i-1} \bar{g}_j),$$

where $Q_i : L_2 \to S_{i,0}$ are L_2-orthoprojections, it is easily observed from (2.6) and Remark 2.2 that the matrix B_X given by

$$((B_X x_N, x_N)) = \|Q_0 \bar{g}_j\|_{L_2}^2 + \sum_{i=1}^{j} 2^{4i} \cdot \|Q_i \bar{g}_j - Q_{i-1} \bar{g}_j\|_{L_2}^2, \qquad g_j \in S_{j,0},$$

is spectrally equivalent to A_N, i.e., $((A_N x_N, x_N)) \approx ((B_X x_N, x_N))$ for all vectors x_N (with equivalence constants independent of the number of refinement levels). Now, repeating the arguments in [B-P-X], [X], [Y4], or [Dö2], we replace B_X^{-1} by a spectrally equivalent and computationally simpler preconditioning matrix C_X indirectly given by

$$x_N \leftrightarrow \bar{g}_j \to \mathscr{C}_X \bar{g}_j = \sum_{i=0}^{j} \sum_{k=0}^{1} \sum_{s'} d_{i,s'}^k \cdot (\bar{g}_j, N_{i,s'}^k) \cdot N_{i,s}^k \leftrightarrow C_X x_N$$

where the scaling factors $d_{i,s'}^k \approx 1$ can be chosen to improve the practical performance of the preconditioner. The final result is an $O(1)$ bound for $\kappa(C_X A_N)$, uniformly in j.

REFERENCES

[B] R. E. BANK, PLTMG *User's Guide.* Edition 5.0, Tech. Report, Department of Mathematics, University of California, San Diego, CA, 1988.

[B-Du-Y] R. E. BANK, T. F. DUPONT, AND H. YSERENTANT, *The hierarchical basis multigrid method,* Numer. Math., 52 (1988), pp. 427–458.

[Be-I-N] O. V. BESOV, V. P. IL'IN, AND S. M. NIKOL'SKIJ, *Integral Representations of Functions and Imbedding Theorems,* Nauka, Moscow, 1975. (In Russian.)

[B-P-X] J. H. BRAMBLE, J. E. PASCIAK, AND J. XU, *Parallel multilevel preconditioners,* Math. Comp., 55 (1990), pp. 1–22.

[Br] S. C. BRENNER, *An optimal-order nonconforming multigrid method for the biharmonic equation,* SIAM J. Numer. Anal., 26 (1989), pp. 1124–1138.

[Ch] C. K. CHUI, *Multivariate Splines,* Society for Industrial and Applied Mathematics, Philadelphia, PA, 1988.

[Ch-He] C. K. CHUI AND T.-X. HE, *Bivariate C^1 quadratic finite elements and vertex splines,* Math. Comp., 54 (1990), pp. 169–187.

[Cia] PH. CIARLET, *The finite element method for elliptic problems,* North-Holland, Amsterdam, New York, Oxford, 1978.

[Da-D-S] W. DAHMEN, R. DEVORE, AND K. SCHERER, *Multidimensional spline approximation,* SIAM J. Numer. Anal., 17 (1980), pp. 380–402.

[D-Pop1] R. DEVORE AND V. A. POPOV, *Free multivariate splines,* Constr. Approx., 3 (1987), pp. 239–248.

[D-Pop2] R. DeVore and V. A. Popov, *Interpolation of Besov spaces*, Trans. Amer. Math. Soc., 305 (1988), pp. 397–414.

[De-L-Y] P. Deuflhard, P. Leinen, and J. Yserentant, *Concepts of an adaptive hierarchical finite element code*, Impact Comput. Sci. Engrng., 1 (1989), pp. 3–35.

[Dö1] W. Dörfler, *Hierarchical bases for elliptic problems*, Preprint 123, SFB 256, Universität Bonn, Bonn, Germany, 1990.

[Dö2] ———, *A note on the preconditioner of Bramble, Pasciak, Xu*, Institut für Angewandte Mathematik, Universität Zürich, Zürich, Switzerland, preprint, 1991.

[Fa] G. Farin, *Triangular Bernstein Bezier patches*, Comput. Aided Geom. Design, 3 (1986), pp. 83–127.

[Me] Y. Meyer, *Ondelettes*, Hermann, Paris, 1990.

[N] S. M. Nikol'skij, *Approximation of Functions of Several Variables and Imbedding Theorems*, 2nd Ed., Nauka i Tekhn. Progress, 1977. (In Russian.)

[Os1] P. Oswald, *On function spaces related to finite element approximation theory*, Z. Anal. Anwendungen, 9 (1990), pp. 43–64.

[Os2] ———, *On estimates for hierarchic basis representations of finite element functions*, Preprint N/89/16 Friedrich Schiller Universität, Jena, Germany, 1989.

[Os3] ———, *On C^1 interpolating hierarchical spline bases*, Preprint N/89/19 Friedrich Schiller Universität, Jena, Germany, 1989.

[Os4] ———, *On the degree of nonlinear spline approximation in Besov–Sobolev spaces*, J. Approx. Theory, 61 (1990), pp. 131–157.

[Os5] ———, *On a hierarchical basis multilevel method with nonconforming P1 elements*, Numer. Math., 62 (1992), pp. 189–212.

[P-Brae] P. Peisker and D. Braess, *A conjugent gradient method and a multigrid algorithm for Morley's finite element approximation of the biharmonic equation*, Numer. Math., 50 (1987), pp. 567–586.

[Po-Sa] M. J. D. Powell and M. A. Sabin, *Piecewise quadratic approximations on triangles*, ACM Trans. Mat. Software, 3 (1977), pp. 316–325.

[Sch] L. L. Schumaker, *On spaces of piecewise polynomials in two variables*, in Approximation Theory and Spline Functions, S. P. Singh et al., eds., Reidel, Boston, 1984, pp. 151–197.

[Si] W. Sickel, *Spline representations of function spaces in Besov–Triebel–Lizorkin spaces on $\mathbb{R}^n$*, Forum Math., 2 (1990), pp. 451–475.

[T] H. Triebel, *Interpolation Theory, Function Spaces, Differential Operators*. Deutscher Verlag für Wissenschaft, Berlin 1978, North-Holland, Amsterdam, New York, Oxford, 1978.

[X] J. Xu, *Theory of Multilevel Methods*, Report AM48, Department of Math., Pennsylvania State University, University Park, PA, 1989.

[Y1] H. Yserentant, *On the multilevel splitting of finite element spaces*, Numer. Math., 49 (1986), pp. 379–412.

[Y2] ———, *Hierarchical bases give conjugate gradient methods a multigrid speed of convergence*, Appl. Math. Comput., 19 (1986), pp. 347–358.

[Y3] ———, *On the multilevel splitting of finite element spaces for indefinite elliptic boundary value problems*, SIAM J. Numer. Anal., 23 (1986), pp. 581–595.

[Y4] ———, *Two preconditioners based on the multilevel splitting of finite element spaces*, Numer. Math., 58 (1990), pp. 163–184.

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